United States Patent Application Publication	20250275823
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	KODAMA; Kazuki et al.

ROBOTIC SURGICAL SYSTEM, CONTROL METHOD FOR ROBOTIC SURGICAL SYSTEM, AND STORAGE MEDIUM

Abstract

A robotic surgical system includes an operation apparatus including an operation unit to receive an operation for a surgical instrument, the operation unit having a coordinate system rotated by a predetermined angle with respect to a coordinate system of a surgical apparatus, and a receiver to receive a change in the predetermined angle by an operator.

Inventors: **KODAMA; Kazuki (Kobe-shi, JP), YAMAMOTO; Daisuke (Kobe-shi, JP)**

Applicant: **KAWASAKI JUKOGYO KABUSHIKI KAISHA** (Kobe-shi, JP)

Family ID: **94637475**

Assignee: **KAWASAKI JUKOGYO KABUSHIKI KAISHA (Kobe-shi, JP)**

Appl. No.: **19/046651**

Filed: **February 06, 2025**

Foreign Application Priority Data

JP	2024-030883	Mar. 01, 2024

Publication Classification

Int. Cl.: **A61B34/37** (20160101); **A61B1/00** (20060101); **A61B34/00** (20160101)

U.S. Cl.:

CPC **A61B34/37** (20160201); **A61B1/00045** (20130101); **A61B1/00055** (20130101); **A61B1/00149** (20130101); **A61B34/25** (20160201);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The priority application number JP2024-030883, Robotic Surgical System and Control Method for Robotic Surgical System, Mar. 1, 2024, KODAMA Kazuki and YAMAMOTO Daisuke, upon which this patent application is based, is hereby incorporated by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to a robotic surgical system, a control method for a robotic surgical system, and a storage medium.

Description of the Background Art

[0003] Conventionally, a robotic surgical system including a robot arm to which a surgical instrument is attached is known. U.S. Pat. No. 6,424,885 discloses a robotic surgical system including a robot arm and a master control console to operate the robot arm. The master control console includes a master controller to be operated by the hand of an operator and a viewer into which the head of the operator is inserted. The angle of the viewer with respect to a horizontal plane is adjustable. In U.S. Pat. No. 6,424,885, the coordinate system of the master controller is automatically adjusted according to the adjusted angle of the viewer. For example, the viewer is adjusted to an angle at which the operator looks down, and the coordinate system of the master controller is adjusted according to the angle of the viewer adjusted to the angle at which the operator looks down.

[0004] However, in U.S. Pat. No. 6,424,885, the coordinate system of the master controller is automatically adjusted according to the angle of the viewer, and thus the operability of the master controller may conceivably be poor depending on the operator. Therefore, it is desired to improve the operability of the master controller.

SUMMARY OF THE INVENTION

[0005] The present disclosure is intended to provide a robotic surgical system, a control method for a robotic surgical system, and a storage medium that each enable improved operability of an operation unit.

[0006] A robotic surgical system according to a first aspect of the present disclosure includes a surgical apparatus including a robot arm to allow a surgical instrument to be attached thereto, an operation apparatus including an operation unit to receive an operation for the surgical instrument, the operation unit having a coordinate system rotated by a predetermined angle with respect to a coordinate system of the surgical apparatus, a controller configured or programmed to perform a control to move the surgical instrument by the robot arm based on the operation received by the operation unit, and a receiver to receive a change in the predetermined angle by an operator.

[0007] In the robotic surgical system according to the first aspect of the present disclosure, the receiver is operable to receive the change in the predetermined angle by the operator. Accordingly, the rotation angle of the coordinate system of the operation unit with respect to the coordinate system of the surgical apparatus can be changed, and thus the coordinate system of the operation unit can be adjusted such that the operator can easily operate the operation unit. Consequently, the operability of the operation unit can be improved.

[0008] A control method for a robotic surgical system according to a second aspect of the present disclosure includes receiving a change by an operator in a predetermined angle of a coordinate system of an operation unit rotated by the predetermined angle with respect to a coordinate system of a surgical apparatus, changing the predetermined angle of the coordinate system of the operation unit based on a received change in the predetermined angle, receiving an operation with the operation unit for a surgical instrument attached to a robot arm in a state in which the coordinate

system of the operation unit has been changed, and moving the surgical instrument by the robot arm based on a received operation.
[0009] As described above, the control method for a robotic surgical system according to the second aspect of the present disclosure includes receiving the change in the predetermined angle by the operator, and changing the predetermined angle of the coordinate system of the operation unit based on the received change in the predetermined angle. Accordingly, the rotation angle of the coordinate system of the operation unit with respect to the coordinate system of the surgical apparatus can be changed, and thus the coordinate system of the operation unit can be adjusted such that the operator can easily operate the operation unit. Consequently, it is possible to provide the control method for a robotic surgical system that enables improved operability of the operation unit.
[0010] A storage medium according to a third aspect of the present disclosure is operable to store a program for a control method for a robotic surgical system, including receiving a change by an operator in a predetermined angle of a coordinate system of an operation unit rotated by the predetermined angle with respect to a coordinate system of a surgical apparatus, changing the predetermined angle of the coordinate system of the operation unit based on a received change in the predetermined angle, receiving an operation with the operation unit for a surgical instrument attached to a robot arm in a state in which the coordinate system of the operation unit has been changed, and moving the surgical instrument by the robot arm based on a received operation.
[0011] As described above, the storage medium according to the third aspect of the present disclosure is operable to store the program for the control method including receiving the change in the predetermined angle by the operator, and changing the predetermined angle of the coordinate system of the operation unit based on the received change in the predetermined angle. Accordingly, the rotation angle of the coordinate system of the operation unit with respect to the coordinate system of the surgical apparatus can be changed, and thus the coordinate system of the operation unit can be adjusted such that the operator can easily operate the operation unit. Consequently, it is possible to provide the storage medium that enables improved operability of the operation unit.
[0012] According to the robotic surgical system, the control method for a robotic surgical system, and the storage medium according to the present disclosure, the operability of the operation unit can be improved.
[0013] The foregoing and other objects, features, aspects and advantages of the present disclosure will become more apparent from the following detailed description of the present disclosure when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0014] FIG. **1** is a diagram showing the configuration of a robotic surgical system according to an embodiment.
[0015] FIG. **2** is a diagram showing a display of a medical cart according to the embodiment.
[0016] FIG. **3** is a diagram showing the configuration of the medical cart according to the embodiment.
[0017] FIG. **4** is a diagram showing the configuration of a robot arm according to the embodiment.
[0018] FIG. **5** is a diagram showing a pair of forceps.
[0019] FIG. **6** is a perspective view showing the configuration of an arm operation unit according to the embodiment.
[0020] FIG. **7** is a diagram for illustrating translational movement of the robot arm.
[0021] FIG. **8** is a diagram for illustrating rotational movement of the robot arm.
[0022] FIG. **9** is a diagram showing an endoscope.

[0023] FIG. **10** is a diagram showing a pivot position setting instrument.
[0024] FIG. **11** is a diagram showing operation units according to the embodiment.
[0025] FIG. **12** is a diagram showing a right-handed wrist according to the embodiment.
[0026] FIG. **13** is a diagram showing a left-handed wrist according to the embodiment.
[0027] FIG. **14** is a perspective view showing foot pedals according to the embodiment.
[0028] FIG. **15** is a control block diagram of the robotic surgical system according to the embodiment.
[0029] FIG. **16** is a control block diagram of the robot arm according to the embodiment.
[0030] FIG. **17** is a control block diagram of a positioner and the medial cart according to the embodiment.
[0031] FIG. **18** is a control block diagram of the operation unit according to the embodiment.
[0032] FIG. **19** is a diagram for illustrating a method for calculating an axis value of each axis.
[0033] FIG. **20** is a diagram showing a base coordinate system and a tool coordinate system with respect to the endoscope.
[0034] FIG. **21** is a diagram showing an endoscope coordinate system.
[0035] FIG. **22** is a diagram showing a display and the endoscope coordinate system.
[0036] FIG. **23** is a diagram showing the coordinate system of the operation unit.
[0037] FIG. **24** is a diagram showing the coordinate system of the operation unit and the endoscope coordinate system.
[0038] FIG. **25** is a diagram showing an operator and the coordinate system of the operation unit.
[0039] FIG. **26** is a diagram showing buttons displayed on a touch panel to change a predetermined angle.
[0040] FIG. **27** is a flowchart for illustrating a control method for the robotic surgical system according to the embodiment.
[0041] FIG. **28** is a diagram for illustrating setting of a virtual plane.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

Configuration of Robotic Surgical System

[0042] The configuration of a robotic surgical system **500** according to this embodiment is now described. The robotic surgical system **500** includes a surgical robot **100**, a remote control apparatus **200**, a vision unit **300**, and an image processing unit **400**. The surgical robot **100** is an example of a surgical apparatus, and the remote control apparatus **200** is an example of an operation apparatus.
[0043] In this specification, as shown in FIG. **4**, the longitudinal direction of a surgical instrument **1** is defined as a Z direction. The distal end side of the surgical instrument **1** is defined as a Z1 side, and the proximal end side of the surgical instrument **1** is defined as a Z2 side. A direction perpendicular to the Z direction is defined as an X direction. A direction perpendicular to the Z direction and the X direction is defined as a Y direction.
[0044] As shown in FIG. **1**, the surgical robot **100** is arranged in an operating room. The remote control apparatus **200** is spaced apart from the surgical robot **100**. The remote control apparatus **200** receives operations for the surgical instrument **1**. Specifically, an operator such as a doctor inputs a command to the remote control apparatus **200** to cause the surgical robot **100** to perform a desired operation. The remote control apparatus **200** transmits the input command to the surgical robot **100**. The surgical robot **100** operates based on the received command. The surgical robot **100** is arranged in the operating room that is a sterilized sterile field.

Configuration of Surgical Robot

[0045] As shown in FIG. **1**, the surgical robot **100** includes a medical cart **10**, a cart positioner operation unit **20**, a positioner **30**, an arm base **40**, a plurality of robot arms **50**, and an arm operation unit **60** provided on each of the robot arms **50**.
[0046] As shown in FIG. **3**, the cart positioner operation unit **20** is supported by a cart positioner operation support **21** at the rear of the medical cart **10**, and the medical cart **10** or the positioner **30**

is moved by operating the cart positioner operation unit **20**. The cart positioner operation unit **20** includes an input **22** and an operation handle **23**. The input **22** receives operations to move the positioner **30**, the arm base **40**, and the plurality of robot arms **50** or change their postures mainly in order to prepare for surgery before the surgery. The medical cart **10** includes the operation handle **23**.

[0047] As shown in FIG. **3**, the input **22** of the medical cart **10** includes a display **22***a*, a joystick **22***b*, an enable switch **22***c*, an error reset button **22***d*, and speakers **22***e*. The display **22***a* is a liquid crystal panel, for example. As shown in FIG. **2**, the display **22***a* displays numbers corresponding to the plurality of robot arms **50**. The display **22***a* also displays the type of surgical instrument **1** attached to each of the plurality of robot arms **50**. A check mark CM indicating that a pivot position PP described below has been set is displayed on the display **22***a*.

[0048] As shown in FIG. **3**, the joystick **22***b* is arranged in the vicinity of or adjacent to the display **22***a* of the input **22** of the medical cart **10**. The positioner **30** is moved three-dimensionally by selecting an operation mode displayed on the display **22***a* and operating the joystick **22***b*.

[0049] The enable switch **22***c* is arranged in the vicinity of or adjacent to the joystick **22***b*. The enable switch **22***c* enables or disables movement of the positioner **30**. When the joystick **22***b* is operated while the enable switch **22***c* is pressed to enable movement of the positioner **30**, the positioner **30** is moved.

[0050] The error reset button **22***d* resets errors in the robotic surgical system **500**. The errors may be abnormal deviation errors, for example. The speakers **22***e* are arranged in pair. The pair of speakers **22***e* are arranged in the vicinity of or adjacent to the location of the positioner **30** in the medical cart **10**.

[0051] The operation handle **23** is arranged in the vicinity of the display **22***a*. The operation handle **23** includes a throttle **23***a* that is gripped and twisted by an operator such as a nurse or a technician to operate movement of the medical cart **10**. Specifically, the operation handle **23** is arranged below the input **22**. As the throttle **23***a* is twisted from the near side to the far side, the medical cart **10** moves forward. As the throttle **23***a* is twisted from the far side to the near side, the medical cart **10** moves rearward. The speed of the medical cart **10** is changed according to a twisting amount of the throttle **23***a*. The operation handle **23** is rotatable to the left and right shown by an R direction, and the medical cart **10** is turned with rotation of the operation handle **23**.

[0052] An enable switch **23***b* for enabling or disabling movement of the medical cart **10** is provided on the operation handle **23** of the cart positioner operation unit **20**. When the throttle **23***a* of the operation handle **23** is operated while the enable switch **23***b* is pressed to enable movement of the medical cart **10**, the medical cart **10** is moved.

[0053] As shown in FIG. **1**, the positioner **30** includes a 7-axis articulated robot, for example. The positioner **30** is arranged on the medical cart **10**. The positioner **30** adjusts the position of the arm base **40**. The positioner **30** moves the position of the arm base **40** three-dimensionally.

[0054] The positioner **30** includes a base **31** and a plurality of links **32** coupled to the base **31**. The plurality of links **32** are coupled to each other by joints **33**.

[0055] The arm base **40** is attached to the distal end of the positioner **30**. The proximal end of each of the plurality of robot arms **50** is attached to the arm base **40**. Each of the plurality of robot arms **50** is able to take a folded and stored posture. The arm base **40** and the plurality of robot arms **50** are covered with sterile drapes and used. Moreover, each of the robot arms **50** supports the surgical instrument **1**.

[0056] A status indicator **41** and an arm status indicator **42** that are shown in FIG. **15** are provided on the arm base **40**. The status indicator **41** indicates the status of the robotic surgical system **500**. The arm status indicator **42** indicates the statuses of the robot arms **50**.

[0057] The plurality of robot arms **50** are arranged. Specifically, four robot arms **50***a*, **50***b*, **50***c*, and **50***d* are arranged. The robot arms **50***a*, **50***b*, **50***c*, and **50***d* have the same or similar configurations as each other.

[0058] As shown in FIG. **4**, each robot arm **50** includes an arm portion **51**, a first link **52**, a second link **53**, and a translation mechanism **54**. The robot arm **50** includes joints JT**1**, JT**2**, JT**3**, JT**4**, JT**5**, JT**6**, JT**7**, and JT**8**. The joints JT**1**, JT**2**, JT**3**, JT**4**, JT**5**, JT**6**, and JT**7** have A1, A2, A3, A4, A5, A6, and A7 axes as rotation axes, respectively. The joint JT**8** has an A8 axis as a linear motion axis. The arm portion **51** includes a base **51***a* and links **51***b*.
[0059] The arm portion **51** includes a 7-axis articulated robot arm. The first link **52** is arranged at the distal end of the arm portion **51**. The arm operation unit **60** described below is attached to the second link **53**. The translation mechanism **54** is arranged between the first link **52** and the second link **53**.
[0060] A holder **55** that holds the surgical instrument **1** is arranged on the second link **53**. The translation mechanism **54** translates the holder **55** to which the surgical instrument **1** is attached between a first position and a second position. The first position refers to an end position on the Z2 side in a range of movement of the holder **55** along the A8 axis by the translation mechanism **54**. The second position refers to an end position on the Z1 side in the range of movement of the holder **55** along the A8 axis by the translation mechanism **54**.
[0061] The surgical instrument **1** is attached to the distal end of each of the plurality of robot arms **50**. The surgical instrument **1** includes a replaceable instrument **2**, an endoscope **3** shown in FIG. **9** to capture an image of a surgical site, and a pivot position setting instrument **4** shown in FIG. **10** to set the pivot position PP, for example. The instrument **2** includes a driven unit **2***a*, a pair of forceps **2***b*, and a shaft **2***c*. The instrument **2** and the endoscope **3** are examples of a surgical instrument.
[0062] As shown in FIG. **1**, the endoscope **3** is attached to the distal end of one of the plurality of robot arms **50**, such as the robot arm **50***c*, and the instrument **2** is attached to the distal end of each of the remaining robot arms **50***a*, **50***b*, and **50***d*, for example. The endoscope **3** is preferably attached to one of two robot arms **50***b* and **50***c* arranged in the center among the four robot arms **50** arranged adjacent to each other.
Configuration of Instrument
[0063] As shown in FIG. **5**, the pair of forceps **2***b* is attached to the distal end of the instrument **2**, for example. At the distal end of the instrument **2**, in addition to the pair of forceps **2***b*, a pair of scissors, a grasper, a needle holder, a microdissector, a stable applier, a tacker, a suction cleaning tool, a snare wire, a clip applier, etc. are arranged as instruments having joints. At the distal end of the instrument **2**, a cutting blade, a cautery probe, a washer, a catheter, a suction orifice, etc. are arranged as instruments having no joint.
[0064] The pair of forceps **2***b* includes a first support **2***d* and a second support **2***e*. The first support **2***d* supports the proximal end sides of jaw members **2***f* and **2***g* such that the jaw members **2***f* and **2***g* are rotatable about an A11 axis. The second support **2***e* supports the proximal end side of the first support **2***d* such that the first support **2***d* is rotatable about an A10 axis. The shaft **2***c* rotates about an A9 axis. The jaw members **2***f* and **2***g* pivot about the A11 axis to open and close.
Configuration of Arm Operation Unit
[0065] As shown in FIG. **6**, the arm operation unit **60** is attached to the robot arm **50** to operate the robot arm **50**. Specifically, the arm operation unit **60** is attached to the second link **53**.
[0066] The arm operation unit **60** includes an enable switch **61**, a joystick **62**, and linear switches **63**, a mode switching button **64**, a mode indicator **65**, a pivot button **66**, and an adjustment button **67**.
[0067] The enable switch **61** is pressed to enable or disable movement of the robot arm **50** in response to the joystick **62** and the linear switches **63**. When the enable switch **61** is pressed by an operator such as a nurse or an assistant grasping the arm operation unit **60**, movement of the surgical instrument **1** by the robot arm **50** is enabled.
[0068] The joystick **62** is an operation tool to control movement of the surgical instrument **1** by the robot arm **50**. The joystick **62** controls a moving direction and a moving speed of the robot arm **50**. The robot arm **50** is moved in accordance with a tilting direction and a tilting angle of the joystick

62.

[0069] The linear switches **63** are switches to move the surgical instrument **1** in the Z direction, which is the longitudinal direction of the surgical instrument **1**. The linear switches **63** include a linear switch **63***a* to move the surgical instrument **1** in a direction in which the surgical instrument **1** is inserted into a patient P, and a linear switch **63***b* to move the surgical instrument **1** in a direction in which the surgical instrument **1** is moved away from the patient P. Both the linear switch **63***a* and the linear switch **63***b* are push-button switches.

[0070] The mode switching button **64** is a push-button switch to switch between a mode for translationally moving the surgical instrument **1** and a mode for rotationally moving the surgical instrument **1**. As shown in FIG. **7**, in the mode for translationally moving the robot arm **50**, the robot arm **50** is moved such that the distal end **1***a* of the surgical instrument **1** is moved in an X-Y plane. As shown in FIG. **8**, in the mode for rotationally moving the robot arm **50**, the robot arm **50** is moved such that the surgical instrument **1** is rotationally moved about a center on the A11 axis of the pair of forceps **2***b* or the distal end of the pair of forceps **2***b* of the instrument **2** as the surgical instrument **1** as a fulcrum when any pivot position PP is not stored in a storage **351**, and the surgical instrument **1** is rotationally moved about the pivot position PP as a fulcrum when the pivot position PP is stored in the storage **351**. In this case, the surgical instrument **1** is rotationally moved with the shaft **1***c* of the surgical instrument **1** inserted into a trocar T. The mode switching button **64** is arranged on a Z-direction side surface of the arm operation unit **60**.

[0071] The mode indicator **65** indicates a switched mode. The mode indicator **65** is on to indicate a rotational movement mode and is off to indicate a translational movement mode. Furthermore, the mode indicator **65** also serves as a pivot position indicator that indicates that the pivot position PP has been set. The mode indicator **65** is arranged on the Z-direction side surface of the arm operation unit **60**.

[0072] The pivot button **66** is a push-button switch to set the pivot position PP that serves as a fulcrum for movement of the surgical instrument **1** attached to the robot arm **50**.

[0073] The adjustment button **67** is a button to optimize the position of the robot arm **50**. After the pivot position PP for the robot arm **50** to which the endoscope **3** has been attached is set, the positions of the other robot arms **50** and the arm base **40** are optimized when the adjustment button **67** is pressed. The adjustment button **67** is a button different from the enable switch **61**.

Remote Control Apparatus

[0074] The remote control apparatus **200** receives operations for the surgical instrument **1**. As shown in FIG. **1**, the remote control apparatus **200** is arranged inside or outside the operating room, for example. The remote control apparatus **200** includes the operation units **110**, foot pedals **120**, a touch panel **130**, a monitor **140**, a support arm **150**, a support bar **160**, an error reset button **161**. The operation units **110** include operation handles for the operator such as a doctor to input a command. The monitor **140** is an example of a display.

Operation Unit

[0075] As shown in FIG. **11**, the operation units **110** each include a handle to operate the surgical instrument **1**. The operation unit **110** receives an operation for the surgical instrument **1**. The operation units **110** include an operation unit **110**L that is located on the left side as viewed from the operator such as a doctor and is to be operated by the left hand of the operator, and an operation unit **110**R that is located on the right side and is to be operated by the right hand of the operator. The operation units **110** include arms **111** and wrists **112**. The operation unit **110**R includes an arm **111**R and a wrist **112**R. The operation unit **110**L includes an arm **111**L and a wrist **112**L.

[0076] The operation units **110** each have joints JT**21**, JT**22**, and JT**23** shown in FIG. **11**, and joints JT**24**, JT**25**, JT**26**, and JT**27** shown in FIGS. **12** and **13**. The rotation axes of the joints JT**21**, JT**22**, JT**23**, JT**24**, JT**25**, JT**26**, and JT**27** are A21, A22, A23, A24, A25, A26, and A27 axes, respectively.

[0077] As shown in FIG. **11**, the arm **111**R includes a link **111***a*, a link **111***b*, and a link **111***c*. The upper end side of the link **111***a* is attached to the remote control apparatus **200** such that the link

111*a* is rotatable about the A21 axis along a vertical direction. The upper end side of the link **111***b* is attached to the lower end side of the link **111***a* such that the link **111***b* is rotatable about the A22 axis along a horizontal direction. A first end side of the link **111***c* is attached to the lower end side of the link **111***b* such that the link **111***c* is rotatable about the A23 axis along the horizontal direction. The wrist **112** is attached to a second end side of the link **111***c* such that the wrist **112** is rotatable about the A24 axis. The link **111***a* is connected to the remote control apparatus **200** by the joint JT**21**. The link **111***a* and the link **111***b* are connected to each other by the joint JT**22**. The link **111***b* and the link **111***c* are connected to each other by the joint JT**23**. The arm **111** supports the wrist **112**. The arm **111**L has the same or similar configuration as the arm **111**R.

[0078] The wrists **112** include the wrist **112**R shown in FIG. **12** and to be operated by the right hand of the operator, and the wrist **112**L shown in FIG. **13** and to be operated by the left hand of the operator. FIG. **12** shows the reference posture of the operation unit **110**R, and FIG. **13** shows the reference posture of the operation unit **110**L. The configuration of the wrist **112**R is the same as or similar to that of the wrist **112**L.

[0079] The wrists **112** each include a link **112***a*, a link **112***b*, a link **112***c*, and a grip **112***d* that is to be operated by the operator such as a doctor. The link **112***a* includes a proximal end connected to the distal end of the arm **111** and rotates about the A24 axis. The link **112***b* includes a proximal end connected to the distal end of the link **112***a* and rotates about the A25 axis with respect to the link **112***a*. The link **112***c* includes a proximal end connected to the distal end of the link **112***b* and a distal end to which the grip **112***d* is connected, and rotates about the A26 axis with respect to the link **112***b*. The grip **112***d* rotates about the A27 axis with respect to the link **112***c*. The link **112***a*, the link **112***b*, and the link **112***c* each have an L shape.

[0080] The wrist **112** includes a pair of grip members **112***e* that are opened and closed by the operator. The grip members **112***e* each include an elongated plate-shaped lever member, and the proximal ends of the pair of grip members **112***e* are rotatably connected to the proximal end of the grip **112***d*. Cylindrical finger insertion portions **112***f* are provided on the grip members **112***e*. The operator inserts their fingers into a pair of finger insertion portions **112***f* to operate the wrist **112**. The proximal ends of the pair of grip members **112***e* are connected to the grip **112***d*, and an angle between the pair of grip members **112***e* is increased or decreased such that an opening angle between the jaw member **2***f* and the jaw member **2***g* is changed. A magnet is provided on one of the grip members **112***e*, and a Hall sensor is provided on the grip **112***d*. When the operator opens and closes the grip members **112***e*, the magnet and the Hall sensor function as an angle detection sensor, and the Hall sensor outputs the opening angle. As the angle detection sensor, the Hall sensor may be provided on the grip member **112***e*, and the magnet may be provided on the grip **112***d*. Alternatively, the magnet or the Hall sensor may be provided as the angle detection sensor on both the grip members **112***e*.

[0081] As shown in FIG. **1**, the monitor **140** is a scope-type display that displays images captured by the endoscope **3**. A notifier **141** is provided on the monitor **140**. The notifier **141** issues an error sound. The support arm **150** supports the monitor **140** so as to align the height of the monitor **140** with the height of the face of the operator such as a doctor. The touch panel **130** is arranged on the support bar **160**. The touch panel **130** receives settings for the robotic surgical system **500**. The head of the operator is detected by a sensor provided in the vicinity of the monitor **140** such that the surgical robot **100** can be operated by the remote control apparatus **200**. The operator operates the operation units **110** and the foot pedals **120** while visually recognizing an affected area on the monitor **140**. Thus, a command is input to the remote control apparatus **200**. The command input to the remote control apparatus **200** is transmitted to the surgical robot **100**.

[0082] The error reset button **161** is arranged on the support bar **160**. The error reset button **161** resets errors in the robotic surgical system **500**. The errors may be abnormal deviation errors, for example.

Foot Pedals

[0083] As shown in FIG. **14**, a plurality of foot pedals **120** are provided to perform functions related to the surgical instrument **1**. The plurality of foot pedals **120** are arranged on a base **121**. The foot pedals **120** include a switching pedal **122**, a clutch pedal **123**, a camera pedal **124**, incision pedals **125**, coagulation pedals **126**, and foot detectors **127**. The switching pedal **122**, the clutch pedal **123**, the camera pedal **124**, the incision pedals **125**, and the coagulation pedals **126** are operated by the foot of the operator. The incision pedals **125** include an incision pedal **125**R for a right robot arm **50**, and an incision pedal **125**L for a left robot arm **50**. The coagulation pedals **126** include a coagulation pedal **126**R for the right robot arm **50** and a coagulation pedal **126**L for the left robot arm **50**.
[0084] The switching pedal **122** switches robot arms **50** to be operated by the operation units **110**. The clutch pedal **123** performs a clutch operation to temporarily disconnect an operation connection between the robot arms **50** and the operation units **110**. While the clutch pedal **123** is being pressed by the operator, operations by the operation units **110** are not transmitted to the robot arms **50**. While the camera pedal **124** is being pressed by the operator, the operation unit **110** can operate a robot arm **50** to which the endoscope **3** is attached. While the incision pedals **125** or the coagulation pedals **126** are being pressed by the operator, an electrosurgical device is activated.
[0085] The foot detectors **127** detect the foot of the operator that operates the foot pedals **120**. The foot detectors **127** are provided for the switching pedal **122**, the clutch pedal **123**, the camera pedal **124**, the incision pedal **125**L, the coagulation pedal **126**L, the incision pedal **125**R, and the coagulation pedal **126**R, and detect the foot that hovers above their corresponding foot pedals **120**. The foot detectors **127** are arranged on the base **121**. The functions of the foot pedals **120** including the camera pedal **124** are not limited to being performed through pedals configured to be pressed by the foot of the operator as in this embodiment, but inputs such as hand switches may be provided on the operation units **110**, and manual operations by the operator may be used, for example.
Vision Unit and Image Processing Unit
[0086] As shown in FIG. **1**, the vision unit **300** and the image processing unit **400** are placed on a cart **210**. The image processing unit **400** processes images captured by the endoscope **3**. A display **220** is arranged on the cart **210**. The display **220** displays images captured by the endoscope **3**. An error reset button **230** and a notifier **240** are arranged on the vision unit **300**. The error reset button **230** resets errors in the robotic surgical system **500**. The errors may be abnormal deviation errors, for example. The notifier **240** issues an error sound.
Configuration of Control System
[0087] As shown in FIG. **15**, the robotic surgical system **500** includes a first control device **310**, an arm controller **320**, a positioner controller **330**, operation controllers **340**, and a second control device **350**. The robotic surgical system **500** also includes a storage **311** connected to the first control device **310** and the storage **351** connected to the second control device **350**. The first control device **310** is an example of a controller. The second control device **350** is an example of a monitoring controller.
[0088] The first control device **310** is accommodated in the medical cart **10** to communicate with the arm controller **320** and the positioner controller **330**, and controls the entire robotic surgical system **500**. Specifically, the first control device **310** communicates with and controls the arm controller **320**, the positioner controller **330**, and the operation controllers **340**. The first control device **310** is connected to the arm controller **320**, the positioner controller **330**, and the operation controllers **340** through a LAN, for example. The first control device **310** is arranged inside the medical cart **10**.
[0089] The arm controller **320** is arranged for each of the plurality of robot arms **50**. That is, the same number of arm controllers **320** as the plurality of robot arms **50** are placed inside the medical cart **10**.
[0090] As shown in FIG. **15**, the input **22** is connected to the first control device **310** through a

LAN, for example. The status indicator **41**, the arm status indicator **42**, the operation handle **23**, the throttle **23***a*, and the joystick **22***b* are connected to the positioner controller **330** via a wire line **360** by means of a communication network that allows information to be shared with each other by using serial communication. Although FIG. **15** shows that the status indicator **41**, the arm status indicator **42**, etc. are all connected to one wire line **360**, in reality, the wire line **360** is arranged for each of the status indicator **41**, the arm status indicator **42**, the operation handle **23**, the throttle **23***a*, and the joystick **22***b*.

[0091] As shown in FIG. **16**, the arm portion **51** includes a plurality of servomotors SM**1**, encoders EN**1**, and speed reducers so as to correspond to the joints JT**1**, JT**2**, JT**3**, JT**4**, JT**5**, JT**6**, and JT**7**. The encoders EN**1** detect rotation angles of the servomotors SM**1**. The speed reducers slow down rotation of the servomotors SM**1** to increase the torques. Inside the medical cart **10**, servo controllers SC**1** that control the servomotors SM**1** are arranged adjacent to the arm controller **320**. In addition, the encoders EN**1** that detect the rotation angles of the servomotors SM**1** are electrically connected to the servo controllers SC**1**.

[0092] Servomotors SM**2** that rotate driven members provided in the driven unit **2***a* of the instrument **2**, encoders EN**2**, and speed reducers are arranged in the second link **53**. The encoders EN**2** detect rotation angles of the servomotors SM**2**. The speed reducers slow down rotation of the servomotors SM**2** to increase the torques. In the medical cart **10**, servo controllers SC**2** are provided to control the servomotors SM**2** to drive the surgical instrument **1**. The encoders EN**2** that detect the rotation angles of the servomotors SM**2** are electrically connected to the servo controllers SC**2**. A plurality of servomotors SM**2**, a plurality of encoders EN**2**, and a plurality of servo controllers SC**2** are arranged.

[0093] The translation mechanism **54** includes a servomotor SM**3** to translationally move the surgical instrument **1**, an encoder EN**3**, and a speed reducer. The encoder EN**3** detects a rotation angle of the servomotor SM**3**. The speed reducer slows down rotation of the servomotor SM**3** to increase the torque. In the medical cart **10**, a servo controller SC**3** is provided to control the servomotor SM**3** to translationally move the surgical instrument **1**. The encoder EN**3** that detects the rotation angle of the servomotor SM**3** is electrically connected to the servo controller SC**3**.

[0094] The first control device **310** generates command values to command the positions of the servomotors SM**1**, SM**2**, and SM**3** based on operations received by the remote control apparatus **200**, and drives the servomotors SM**1**, SM**2**, and SM**3** based on the command values. The first control device **310** detects an abnormal deviation error when differences between the command values and the positions of the servomotors SM**1**, SM**2**, and SM**3** detected by sensors exceed an allowable range.

[0095] As shown in FIG. **17**, a plurality of servomotors SM**4**, a plurality of encoders EN**4**, and a plurality of speed reducers are provided in the positioner **30** so as to correspond to a plurality of joints **33** of the positioner **30**. The encoders EN**4** detect rotation angles of the servomotors SM**4**. The speed reducers slow down rotation of the servomotors SM**4** to increase the torques.

[0096] The medical cart **10** includes wheels including front wheels as drive wheels and rear wheels that are steered by the operation handle **23**. The rear wheels are arranged closer to the operation handle **23** than the front wheels. The medical cart **10** includes servomotors SM**5** to drive a plurality of front wheels of the medical cart **10**, encoders EN**5**, speed reducers, and brakes BRK. The speed reducers slow down rotation of the servomotors SM**5** to increase the torques. A potentiometer P**1** shown in FIG. **3** is provided on the operation handle **23** of the medical cart **10**, and the servomotors SM**5** of the front wheels are driven based on a rotation angle detected by the potentiometer P**1** according to the twist of the throttle **23***a*. Rear wheels of the medical cart **10** are of the dual wheel type, and the rear wheels are steered based on rightward-leftward rotation of the operation handle **23**. Furthermore, a potentiometer P**2** shown in FIG. **3** is provided on a rotation axis of the operation handle **23** of the medical cart **10**, and servomotors SM**6**, encoders EN**6**, and speed reducers are provided on the rear wheels of the medical cart **10**. The speed reducers slow down rotation of the

servomotors SM**6** to increase the torques. The servomotors SM**6** are driven based on a rotation angle detected by the potentiometer P**2** according to rightward-leftward rotation of the operation handle **23**. That is, steering of the rear wheels by the rightward-leftward rotation of the operation handle **23** is power-assisted by the servomotors SM**6**.
[0097] The front wheels of the medical cart **10** are driven such that the medical cart **10** moves in a forward-rearward direction. Furthermore, the operation handle **23** of the medical cart **10** is rotated such that the rear wheels are steered, and the medical cart **10** turns in a right-left direction.
[0098] As shown in FIG. **17**, in the medical cart **10**, servo controllers SC**4** are provided to control the servomotors SM**4** to move the positioner **30**. The encoders EN**4** that detect the rotation angles of the servomotors SM**4** are electrically connected to the servo controllers SC**4**. In the medical cart **10**, servo controllers SC**5** are provided to control the servomotors SM**5** to drive the front wheels of the medical cart **10**. The encoders EN**5** that detect the rotation angles of the servomotors SM**5** are electrically connected to the servo controllers SC**5**. In the medical cart **10**, servo controllers SC**6** are provided to control the servomotors SM**6** to power-assist steering of the rear wheels of the medical cart **10**. The encoders EN**6** that detect the rotation angles of the servomotors SM**6** are electrically connected to the servo controllers SC**6**.
[0099] As shown in FIGS. **16** and **17**, the joints JT**1**, JT**2**, JT**3**, JT**4**, JT**5**, JT**6**, and JT**7** of the arm portion **51**, and the joints **33** of the positioner **30** include brakes BRK. Furthermore, the front wheels of the medical cart **10**, the arm base **40**, and the translation mechanism **54** include brakes BRK. The arm controller **320** unidirectionally transmits control signals to the brakes BRK of the joints JT**1**, JT**2**, JT**3**, JT**4**, JT**5**, JT**6**, and JT**7** of the arm portion **51**, and the translation mechanism **54**. The control signals are signals for turning on/off the brakes BRK. The signals for turning on the brakes BRK include signals for maintaining the brakes BRK in an enabled state. The same applies to control signals from the positioner controller **330** to the brakes BRK of the joints **33** of the positioner **30** and the arm base **40**. On startup, all the brakes BRK of the arm base **40**, the arm portion **51**, and the translation mechanism **54** are turned off but the servomotors SM are driven against gravity to maintain the postures of the robot arm **50** and the arm base **40**. When an error occurs in the robotic surgical system **500**, the brakes BRK of the arm base **40**, the arm portion **51** and the translation mechanism **54** are turned on. When the error in the robotic surgical system **500** is reset, the brakes BRK of the arm base **40**, the arm portion **51**, and the translation mechanism **54** are turned off. When a shutdown operation is performed in the robotic surgical system **500**, the brakes BRK of the arm base **40**, the arm portion **51**, and the translation mechanism **54** are turned on. The brakes BRK of the front wheels of the medical cart **10** are constantly turned on, and the brakes BRK are deactivated only while the enable switch **23***b* of the medical cart **10** is being pressed. The brakes BRK of the joints **33** of the positioner **30** are constantly turned on, and the brakes BRK are deactivated only while the enable switch **22***c* of the medical cart **10** is being pressed.
[0100] As shown in FIG. **18**, servomotors SM**7***a*, SM**7***b*, SM**7***c*, SM**7***d*, SM**7***e*, SM**7***f*, and SM**7***g* are arranged on the joints JT**21**, JT**22**, JT**23**, JT**24**, JT**25**, JT**26**, and JT**27** of the operation unit **110**, respectively. The servomotor SM**7***a* rotates the link **111***a* about the A21 axis. The servomotor SM**7***b* rotates the link **111***b* about the A22 axis. The servomotor SM**7***c* rotates the link **111***c* about the A23 axis. The servomotor SM**7***d* rotates the link **112***a* about the A24 axis. The servomotor SM**7***e* rotates the link **112***b* about the A25 axis. The servomotor SM**7***f* rotates the link **112***c* about the A26 axis. The servomotor SM**7***g* rotates the grip **112***d* about the A27 axis. Servo controllers SC**7***a*, SC**7***b*, SC**7***c*, SC**7***d*, SC**7***e*, SC**7***f*, and SC**7***g* are provided to control the servomotors. Encoders EN**7***a*, EN**7***b*, EN**7***c*, EN**7***d*, EN**7***e*, EN**7***f*, and EN**7***g* are electrically connected to the servo controllers to detect rotation angles of the servomotors. The servomotors, the servo controllers, and the encoders are provided in each of the operation unit **110**L and the operation unit **110**R.
[0101] The first control device **310** controls the servomotors via the operation controllers **340** to generate torques that counteract gravitational torques generated on rotation axes of the servomotors

according to the postures of the operation units **110**. Thus, the operator can operate the operation units **110** with a relatively small force.

[0102] As shown in FIG. **15**, the first control device **310** controls the robot arm **50** based on an operation received by the arm operation unit **60**. For example, the first control device **310** controls the robot arm **50** based on an operation received by the joystick **62** of the arm operation unit **60**. Specifically, the arm controller **320** outputs an input signal input from the joystick **62** to the first control device **310**. The first control device **310** generates position commands based on the received input signal and the rotation angles detected by the encoders EN**1**, and outputs the position commands to the servo controllers SC**1** via the arm controller **320**. The servo controllers SC**1** generate current commands based on the position commands input from the arm controller **320** and the rotation angles detected by the encoders EN**1**, and output the current commands to the servomotors SM**1**. Thus, the robot arm **50** is moved according to an operation command input to the joystick **62**.

[0103] The first control device **310** controls the robot arm **50** based on an input signal from either linear switch **63** of the arm operation unit **60**. Specifically, the arm controller **320** outputs the input signal input from the linear switch **63** to the first control device **310**. The first control device **310** generates a position command(s) based on the received input signal and the rotation angle(s) detected by the encoders EN**1** or the encoder EN**3**, and outputs the position command(s) to the servo controllers SC**1** or the servo controller SC**3** via the arm controller **320**. The servo controllers SC**1** or the servo controller SC**3** generates a current command(s) based on the position command(s) input from the arm controller **320** and the rotation angle(s) detected by the encoders EN**1** or the encoder EN**3**, and outputs the current command(s) to the servomotors SM**1** or the servomotor SM**3**. Thus, the robot arm **50** is moved according to an operation command input to the linear switch **63**.

[0104] The positioner controller **330** is arranged in the medical cart **10**. The positioner controller **330** controls the positioner **30** and the medical cart **10**. The servomotors SM**4**, the encoders EN**4**, and the speed reducers are provided in the positioner **30** so as to correspond to the plurality of joints **33** of the positioner **30**. The servo controllers SC**4** are provided in the medical cart **10** to control the servomotors SM**4** of the positioner **30**. The servomotors SM**5** and SM**6** that drive the plurality of front wheels of the medical cart **10**, the encoders EN**5** and EN**6**, the speed reducers, the servo controllers SC**5** and SC**6**, and the brakes BRK are provided in the medical cart **10**.

[0105] The operation controllers **340** are arranged in a main body of the remote control apparatus **200**. The operation controllers **340** control the operation units **110**. The operation controllers **340** are provided so as to correspond to the left-handed operation unit **110**L and the right-handed operation unit **110**R, respectively, as shown in FIG. **15**.

[0106] As shown in FIG. **15**, the vision unit **300** and the image processing unit **400** are connected to the first control device **310** through a LAN, for example. The display **220** is connected to the vision unit **300**.

Control Operation of First Control Device

[0107] A control performed by the first control device **310** when the operation unit **110** receives an operation by the operator is now described.

[0108] As shown in FIG. **19**, the first control device **310** presets an operation unit matrix for the operation unit **110**. The operation unit matrix is a homogeneous transformation matrix of a 4×4 matrix. The operation unit matrix represents the result of forward kinematics calculation for the position and posture of the operation unit **110**. The operation unit matrix is set based on the coordinate system C**1** of the operation unit **110**. The coordinate system C**1** of the operation unit **110** is described below. Next, the operation by the operator is received by the operation unit **110**. Thus, a target matrix that is a target corresponding to the received operation is generated. The target matrix is also a homogeneous transformation matrix. The target matrix represents target values of the position and posture of the surgical instrument **1**. The target matrix is set based on the coordinate system of the surgical robot **100**. The homogeneous transformation matrix includes a translation component for translation of the surgical instrument **1** and a rotation component for rotation of the surgical instrument **1**. Specifically, the first control device **310** calculates a difference between the current position of the surgical instrument **1** and a target position received by the operation unit **110**. The position corresponds to the translation component of the homogeneous transformation matrix. The first control device **310** calculates a difference between the current posture of the surgical instrument **1** and a target posture received by the operation unit **110**. The posture corresponds to the rotation component of the homogeneous transformation matrix. The first control device **310** calculates the target matrix based on the calculated difference value. Next, the first control device **310** performs an inverse kinematics calculation on the updated homogeneous transformation matrix. The first control device **310** calculates axis values of joint axes and linear motion axes for the robot arm **50** and the surgical instrument **1** by the inverse kinematics calculation. Thus, the positions and postures of the robot arm **50** and the surgical instrument **1** are changed to conform to the received operation.

Coordinate Systems of Robotic Surgical System

[0109] The coordinate systems set in the robotic surgical system **500** are now described. First, the coordinate system of the surgical robot **100** is described. As shown in FIGS. **20** and **21**, the coordinate system of the surgical robot **100** includes a base coordinate system C**11**, a tool coordinate system C**12**, and an endoscope coordinate system C**13**. The three axes of the base coordinate system C**11** are defined as an Xa axis, a Ya axis, and a Za axis. The three axes of the tool coordinate system C**12** are defined as an Xb axis, a Yb axis, and a Zb axis. The three axes of the endoscope coordinate system C**13** are defined as an Xc axis, a Yc axis, and a Zc axis. The origin of the base coordinate system C**11** is an intersection of the rotation axis of the proximal end of the positioner **30** with the mounting surface of the medical cart **10** mounted on the positioner **30**. The origin of the tool coordinate system C**12** is set to the clevis position of the pair of forceps **2***b* when the surgical instrument **1** is an instrument **2**, and is set to the distal end position of the endoscope **3** when the surgical instrument **1** is an endoscope **3**. The Zb axis of the tool coordinate system C**12** is along a direction in which the shaft **2***c* of the instrument **2** or a shaft **3***c* of the endoscope **3** extends. The endoscope coordinate system C**13** is obtained by rotating the tool coordinate system C**12** by 180 degrees around the Zb axis. When the surgical instrument **1** is an endoscope **3**, as shown in FIG. **21**, a direction of view of the endoscope **3** may intersect with the direction in which the shaft **3***c* of the endoscope **3** extends. In such a case, the endoscope coordinate system C**13** is obtained by rotating the tool coordinate system C**12** by 180 degrees around the Zb axis and further rotating it around the Xb axis by the intersection angle of the direction of view of the endoscope **3**.

[0110] In this embodiment, an image captured by the endoscope **3** is displayed on the display **220** of the cart **210** shown in FIG. **22**. The image displayed on the display **220** is displayed based on the endoscope coordinate system C**13** of the endoscope **3**. In this embodiment, the image captured by the endoscope **3** is also displayed on the monitor **140** of the remote control apparatus **200**. The image displayed on the monitor **140** is displayed based on the endoscope coordinate system C**13** of the endoscope **3**.

[0111] Next, the coordinate system C**1** of the operation unit **110** is described. As shown in FIG. **23**, a point at which the A24 axis, the A25 axis, the A26 axis, and the A27 axis of the operation unit **110** intersect with each other is called a gimbal point GP. The origin of the coordinate system C**1** of the operation unit **110** is the gimbal point GP. The coordinate system C**1** of the operation unit **110** is set individually for the right-handed operation unit **110**R and the left-handed operation unit **110**L. The three axes of the coordinate system C**1** of the operation unit **110** are an Xd axis, a Yd axis, and a Zd axis.

[0112] In this embodiment, as shown in FIG. **24**, the coordinate system C**1** of the operation unit **110** is rotated by a predetermined angle θ with respect to the coordinate system of the surgical robot **100**. Specifically, the coordinate system C**1** of the operation unit **110** is the same as a coordinate system obtained by rotating the endoscope coordinate system C**13** by 180 degrees around the Zc axis and rotating it by −90 degrees+the predetermined angle θ around the Xc axis. That is, as compared with the endoscope coordinate system C**13** indicated by dotted arrows in FIG. **24**, the Xd axis of the coordinate system C**1** of the operation unit **110** and the Xc axis of the endoscope coordinate system C**13** have opposite positive and negative directions. Furthermore, the Zd axis of the coordinate system C**1** of the operation unit **110** is rotated by the predetermined angle θ around the Xc axis with respect to the Zc axis of the endoscope coordinate system C**13**. Moreover, the Xd axis of the coordinate system C**1** of the operation unit **110** is along the right-left direction of the operator who operates the remote control apparatus **200**. The Yd axis of the coordinate system C**1** of the operation unit **110** is along the forward-rearward direction of the operator who operates the remote control apparatus **200**. The endoscope coordinate system C**13** is an example of a coordinate system of a surgical apparatus.

[0113] In this embodiment, as shown in FIG. **25**, the predetermined angle θ of the coordinate system C**1** of the operation unit **110** is an angle by which the coordinate system C**1** of the operation unit **110** is rotated in a direction away from the operator with respect to a line L perpendicular to a surface on which the remote control apparatus **200** is placed. In other words, the coordinate system C**1** of the operation unit **110** is rotated by the predetermined angle θ toward the far side rather than the near side as viewed from the operator.

[0114] In this embodiment, as shown in FIG. **23**, the touch panel **130** of the remote control apparatus **200** receives a change in the predetermined angle θ. The predetermined angle θ can be changed in a range from 0 degrees to 50 degrees, for example. The touch panel **130** is arranged on the support bar **160**. The touch panel **130** is arranged at a location corresponding to a location between the right-handed operation unit **110**R and the left-handed operation unit **110**L. In other words, the touch panel **130** is arranged in front of the operator. The touch panel **130** is an example of a receiver.

[0115] In this embodiment, as shown in FIG. **26**, the touch panel **130** receives a change in the predetermined angle θ in fixed angular increments. For example, buttons **131** are provided on the touch panel **130** to change the predetermined angle θ. The buttons **131** include a button **131***a* to increase the predetermined angle θ and a button **131***b* to decrease the predetermined angle θ. Each time the button **131***a* is pressed once, the predetermined angle θ increases by one degree, for example. Each time the button **131***b* is pressed once, the predetermined angle θ decreases by one degree, for example. The variable range of the predetermined angle θ may be set. In such a case, a change in the predetermined angle θ is received within the variable range.

[0116] In this embodiment, the storage **311** shown in FIG. **15** stores the predetermined angle θ received for each operator. The first control device **310** retrieves the predetermined angle θ received for each operator and stored in the storage **311**, and sets the coordinate system C**1** of the operation unit **110** based on the retrieved predetermined angle θ. For example, an identifier such as an ID or a name for identifying the operator is registered by the operator operating the touch panel **130**. After the operator operates the touch panel **130** to complete the change of the predetermined angle θ, the changed predetermined angle θ is stored in the storage **311** in association with the identifier. When the operator operates the touch panel **130** to select an identifier, the first control device **310** retrieves the predetermined angle θ associated with the identifier from the storage **311**, and sets the coordinate system C**1** of the operation unit **110** based on the retrieved predetermined angle θ.

[0117] In this embodiment, as shown in FIG. **23**, the rotation angle of the monitor **140** of the remote control apparatus **200** is adjusted around an axis A along the right-left direction of the operator who operates the remote control apparatus **200**. Specifically, the operator manually rotates the monitor **140** around the axis A with respect to the support arm **150**. The touch panel **130** of the remote control apparatus **200** receives a change in the predetermined angle θ independently of the adjustment of the rotation angle of the monitor **140**. That is, the rotation angle of the monitor **140** and the predetermined angle θ are not linked to each other and are adjusted separately. Therefore, it is possible to change the predetermined angle θ such that the rotation angle of the monitor **140** and the predetermined angle θ are equal to each other, and it is also possible to change the predetermined angle θ such that the rotation angle of the monitor **140** and the predetermined angle θ are different from each other.

Control Method for Robotic Surgical System

[0118] A control method for the robotic surgical system **500** is now described.

[0119] As shown in FIG. **27**, in step S**1**, the robotic surgical system **500** is started up. The coordinate system C**1** of the operation unit **110** is rotated by the predetermined angle θ with respect to the coordinate system of the surgical robot **100**, and the predetermined angle θ is a predetermined value. The predetermined value is, for example, 15 degrees and is stored in the storage **311**. The first control device **310** retrieves the predetermined value stored in the storage **311** and sets the coordinate system C**1** of the operation unit **110**.

[0120] In step S**2**, when the operator operates the touch panel **130**, the first control device **310** receives a change in the predetermined angle θ by the operator.

[0121] In step S**3**, the first control device **310** changes the coordinate system C**1** of the operation unit **110** based on the received change in the predetermined angle θ.

[0122] In step S**4**, the first control device **310** receives an operation with the operation unit **110** for the surgical instrument **1** attached to the robot arm **50** in a state in which the coordinate system C**1** of the operation unit **110** has been changed. The operation of the operation unit **110** is performed by the operator.

[0123] An operation to move the endoscope **3** is described. While the camera pedal **124** of the foot pedals **120** shown in FIG. **14** is being pressed by the operator, the robot arm **50** to which the endoscope **3** has been attached can be operated with the operation unit **110**. The endoscope **3** is moved by the operator operating both the right-handed operation unit **110**R and the left-handed operation unit **110**L. Specifically, as shown in FIG. **28**, the first control device **310** defines a virtual plane SF based on the coordinate system C**1** of the operation unit **110**. A method for setting the virtual plane SF is as follows. The gimbal points GP of the operation units **110**R and **110**L are set as GPR and GPL, respectively. The midpoint of a line segment connecting the gimbal point GPR of the operation unit **110**R to the gimbal point GPL of the operation unit **110**L is set as CP. A point, the Y coordinate of which is the same as that of the midpoint CP and the X and Z coordinates of which are the same as those of a midpoint between the midpoint CP and the gimbal point GPR, is set as CP**1**. A straight line connecting the midpoint CP to the point CP**1** is set as a diameter, and the intersection of a circle CL on a plane that passes through the same Y coordinate as the midpoint CP and the point CP**1** with a line perpendicular to the straight line connecting the midpoint CP to the point CP**1** is set as CP**2**. The virtual plane SF is a plane including CP, CP**1**, and CP**2**. When the predetermined angle θ of the coordinate system C**1** of the operation unit **110** is changed, the rotation angle of the virtual plane SF around the Xd axis is also changed. The first control device **310** sets the movement amount of the midpoint CP in the previous control cycle and the current control cycle as the movement amount of the distal end of the endoscope **3**. Specifically, the first control device **310** calculates a homogeneous transformation matrix that is a target for the position of the endoscope **3** in the current control cycle by adding the movement amount of the midpoint CP to a homogeneous transformation matrix that represents the position of the endoscope **3** in the previous control cycle. The first control device **310** performs an inverse kinematics calculation on the updated homogeneous transformation matrix. The first control device **310** calculates the axis values of the joint axes and the linear motion axes for the robot arm **50** and the surgical instrument **1** by the inverse kinematics calculation.

[0124] An operation to move the instrument **2** is described. The operator operates either the right-handed operation unit **110**R or the left-handed operation unit **110**L such that the robot arm **50** to which the instrument **2** has been attached can be operated. The first control device **310** calculates a homogeneous transformation matrix that is a target for the position of the instrument **2** in the current control cycle by adding the movement amount of the distal end of the instrument **2** to a homogeneous transformation matrix that represents the distal end position of the instrument **2** in the previous control cycle. The first control device **310** performs an inverse kinematics calculation on the updated homogeneous transformation matrix. The first control device **310** calculates the axis values of the joint axes and the linear motion axes for the robot arm **50** and the surgical instrument **1** by the inverse kinematics calculation. When the instrument **2** is moved, the coordinate system C**1** of the operation unit **110** becomes the same as a coordinate system obtained by rotating the endoscope coordinate system C**13** by 180 degrees around the Zc axis and rotating it by −90 degrees+the predetermined angle θ around the Xc axis as in the case when the endoscope **3** is moved.

[0125] In step S**5**, the first control device **310** moves the surgical instrument **1** by the robot arm **50** based on the received operation. Until the change of the predetermined angle θ is performed again, the operation in step S**5** is continued based on the coordinate system C**1** of the operation unit **110**, the predetermined angle θ of which has been once changed.

[0126] In this embodiment, the second control device **350** monitors the command values of the first control device **310** for the robot arm **50** and the surgical instrument **1**, and the actual axis values of the robot arm **50** and the surgical instrument **1**. Specifically, during the operation period of step S**5**, the second control devices **350** monitors the command values of the first control device **310** and the actual axis values of the robot arm **50** and the surgical instrument **1** transmitted from the encoders E**1**, E**2**, and E**3**. The axis values are the rotation angles of the servomotors SM for the joint axes and the linear motion axes. When differences between the command values of the first control device **310** and the actual axis values of the robot arm **50** and the surgical instrument **1** exceed a predetermined threshold, the second control device **350** performs a control to issue a warning. For example, an alarm sound is generated from the speakers **22***e* of the medical cart **10** shown in FIG. **3**.

Advantages of This Embodiment

[0127] The touch panel **130** is operable to receive a change in the predetermined angle θ by the operator. Accordingly, the rotation angle of the coordinate system C**1** of the operation unit **110** with respect to the coordinate system of the surgical robot **100** can be changed, and thus the coordinate system C**1** of the operation unit **110** can be adjusted such that the operator can easily operate the operation unit **110**. Consequently, the operability of the operation unit **110** can be improved.

[0128] The predetermined angle θ of the coordinate system C**1** of the operation unit **110** is an angle by which the coordinate system C**1** of the operation unit **110** is rotated in the direction away from the operator with respect to the line L perpendicular to the surface on which the remote control apparatus **200** is placed. Accordingly, the operator operates the operation unit **110** while looking down during surgery, and thus the angle of the coordinate system C**1** of the operation unit **110** can be adjusted in a direction in which the operator looks down.

[0129] The touch panel **130** on the remote control apparatus **200** to receive settings for the robotic surgical system **500** is operable to receive a change in the predetermined angle θ. Accordingly, the change in the predetermined angle θ is received by the touch panel **130** provided on the remote control apparatus **200** in advance, and thus the complexity of the configuration of the robotic surgical system **500** can be reduced or prevented unlike a case in which a separate touch panel **130** is provided to receive a change in the predetermined angle θ.

[0130] The touch panel **130** is operable to receive a change in the predetermined angle θ in fixed angular increments. Accordingly, the amplitude of change in the predetermined angle θ is constant, and thus the possibility that the degree of change in the predetermined angle θ varies with each single operation by the operator can be reduced or prevented.

[0131] The surgical instrument **1** includes the endoscope **3**. The robotic surgical system **500** includes the monitor **140** and the display **220** to display an image captured by the endoscope **3** in accordance with the endoscope coordinate system C**13**. Accordingly, the operator who operates the remote control apparatus **200** can appropriately adjust the coordinate system C**1** of the operation unit **110** such that the operator can easily operate the operation unit **110** while visually checking the image captured by the endoscope **3** and displayed on the monitor **140**. In addition, an assistant or the like other than the operator who operates the remote control apparatus **200** can visually check, on the display **220**, the same image captured by the endoscope **3** that the operator visually checks on the monitor **140**.

[0132] The robotic surgical system **500** includes the storage **311** to store the predetermined angle θ received for each operator. The first control device **310** is configured or programmed to retrieve the predetermined angle θ received for each operator and stored in the storage **311**, and set the coordinate system C**1** of the operation unit **110** based on the retrieved predetermined angle θ. Accordingly, the operator does not need to change the predetermined angle θ every time the robotic surgical system **500** is started up, and thus the need for the operator to change the predetermined angle θ can be eliminated.

[0133] The surgical instrument **1** includes the endoscope **3**. The remote control apparatus **200** includes the monitor **140** to display an image captured by the endoscope **3** and allow the head of the operator to be inserted thereinto. The rotation angle of the monitor **140** is adjusted around the Xd axis along the right-left direction of the operator who operates the remote control apparatus **200**, and the touch panel **130** is operable to receive a change in the predetermined angle θ independently of the adjustment of the rotation angle of the monitor **140**. Accordingly, the predetermined angle θ can be changed to a desired angle such that each operator can more easily operate the operation unit **110** than when the predetermined angle θ of the coordinate system C**1** of the operation unit **110** is automatically changed in conjunction with the change of the rotation angle of the monitor **140**.

[0134] The second control device **350** is configured or programmed to monitor the command values of the first control device **310** for the robot arm **50** and the surgical instrument **1**, and the actual axis values of the robot arm **50** and the surgical instrument **1**. The second control device **350** is configured or programmed to perform a control to issue a warning when the differences between the command values of the first control device **310** and the actual axis values of the robot arm **50** and the surgical instrument **1** exceed the predetermined threshold. Accordingly, it is possible to reduce or prevent the possibility that the operator continues the operation in a state in which the command values of the first control device **310** are different from the actual axis values of the robot arm **50** and the surgical instrument **1** even when the predetermined angle θ of the coordinate system C**1** of the operation unit **110** is changed.

Modified Examples

[0135] The embodiment disclosed this time must be considered as illustrative in all points and not restrictive. The scope of the present disclosure is not shown by the above description of the embodiment but by the scope of claims for patent, and all modifications or modified examples within the meaning and scope equivalent to the scope of claims for patent are further included.

[0136] While the predetermined angle θ of the coordinate system C**1** of the operation unit **110** is an angle by which the coordinate system C**1** of the operation unit **110** is rotated in the direction away from the operator with respect to the line L perpendicular to the surface on which the remote control apparatus **200** is placed in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, the predetermined angle θ of the coordinate system C**1** of the operation unit **110** may alternatively be an angle by which the coordinate system C**1** of the operation unit **110** is rotated in a direction toward the operator with respect to the line L perpendicular to the surface on which the remote control apparatus **200** is placed.

[0137] While a change in the predetermined angle θ is received by the touch panel **130** of the remote control apparatus **200** in the aforementioned embodiment, the present disclosure is not limited to this. For example, a change in the predetermined angle θ may alternatively be received

by an input other than the touch panel **130**, such as a keyboard.
[0138] While the touch panel **130** receives a change in the predetermined angle θ in fixed angular increments in the aforementioned embodiment, the present disclosure is not limited to this. For example, the receiver that receives a change in the predetermined angle θ may alternatively be a rotary dial or slide switch, and a change in the predetermined angle θ may alternatively be received as a continuous value rather than in fixed angular increments.
[0139] While the coordinate system C**1** of the operation unit **110** is rotated by the predetermined angle θ with respect to the endoscope coordinate system C**13** in the aforementioned embodiment, the present disclosure is not limited to this. For example, the coordinate system C**1** of the operation unit **110** may alternatively be rotated by the predetermined angle θ with respect to the coordinate system of the surgical robot **100** other than the endoscope coordinate system C**13**.
[0140] While the coordinate system C**1** of the operation unit **110** is set based on the predetermined angle θ received for each operator and stored in the storage **311** in the aforementioned embodiment, the present disclosure is not limited to this. For example, the changed predetermined angle θ may not be stored in the storage **311**.
[0141] While the second control device **350** monitors the command values of the first control device **310** and the actual axis values of the robot arm **50** and the surgical instrument **1**, and performs a control to issue a warning when the differences between the command values of the first control device **310** and the axis values exceed the predetermined threshold in the aforementioned embodiment, the present disclosure is not limited to this. For example, the first control device **310** itself may alternatively monitor the command values of the first control device **310** and the actual axis values of the robot arm **50** and the surgical instrument **1**, and perform a control to issue a warning when the differences between the command values of the first control device **310** and the axis values exceed the predetermined threshold.
[0142] While the remote control apparatus **200** includes two operation units **110** to operate two robot arms **50** in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, the remote control apparatus **200** may alternatively include only one operation unit **110** to operate one robot arm **50**.
[0143] While as the controller according to the present disclosure, the first control device **310** arranged in the medical cart **10** is applied in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, a controller other than the first control device **310** may alternatively be applied as the controller according to the present disclosure.
[0144] While the touch panel **130** of the remote control apparatus **200** receives a change in the predetermined angle θ independently of adjustment of the rotation angle of the monitor **140** in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, the predetermined angle θ may alternatively be changed in conjunction with rotation of the monitor **140**. The touch panel **130** may alternatively receive a further change in the predetermined angle θ that has been changed in conjunction with rotation of the monitor **140**. Accordingly, the predetermined angle θ is automatically changed in conjunction with rotation of the monitor **140**, and thus it is possible to save the operator time and effort. Furthermore, the touch panel **130** allows fine adjustment of the predetermined angle θ that has been changed in conjunction with rotation of the monitor **140**.
[0145] While four robot arms **50** are provided in the aforementioned embodiment, the present disclosure is not limited to this. In the present disclosure, the number of robot arms **50** may alternatively be any number as long as there is at least one.
[0146] While each of the arm portion **51** and the positioner **30** includes a 7-axis articulated robot in the aforementioned embodiment, the present disclosure is not limited to this. For example, each of the arm portion **51** and the positioner **30** may alternatively include an articulated robot having an axis configuration other than the 7-axis articulated robot. The axis configuration other than the 7-axis articulated robot refers to six axes or eight axes, for example.

[0147] While the surgical robot **100** includes the medical cart **10**, the positioner **30**, the arm base **40**, and the robot arms **50** in the aforementioned embodiment, the present disclosure is not limited to this. For example, the surgical robot **100** may not include the medical cart **10**, the positioner **30**, or the arm base **40**, but may include only the robot arms **50**.

[0148] The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry that includes general purpose processors, special purpose processors, integrated circuits, application specific integrated circuits (ASICs), conventional circuitry and/or combinations thereof that are configured or programmed to perform the disclosed functionality. Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. In the present disclosure, the circuitry, units, or means are hardware that carries out the recited functionality or hardware that is programmed to perform the recited functionality. The hardware may be hardware disclosed herein or other known hardware that is programmed or configured to carry out the recited functionality. When the hardware is a processor that may be considered a type of circuitry, the circuitry, means, or units are a combination of hardware and software, and the software is used to configure the hardware and/or processor.

ASPECTS

[0149] It will be appreciated by those skilled in the art that the exemplary embodiments described above are specific examples of the following aspects.

Item 1

[0150] A robotic surgical system comprising: [0151] a surgical apparatus including a robot arm to allow a surgical instrument to be attached thereto; [0152] an operation apparatus including an operation unit to receive an operation for the surgical instrument, the operation unit having a coordinate system rotated by a predetermined angle with respect to a coordinate system of the surgical apparatus; [0153] a controller configured or programmed to perform a control to move the surgical instrument by the robot arm based on the operation received by the operation unit; and [0154] a receiver to receive a change in the predetermined angle by an operator.

Item 2

[0155] The robotic surgical system according to item 1, wherein the predetermined angle is an angle by which the coordinate system of the operation unit is rotated in a direction away from the operator with respect to a line perpendicular to a surface on which the operation apparatus is placed.

Item 3

[0156] The robotic surgical system according to item 1 or 2, wherein the receiver includes a touch panel on the operation apparatus to receive a setting for the robotic surgical system.

Item 4

[0157] The robotic surgical system according to any one of items 1 to 3, wherein the receiver is operable to receive the change in the predetermined angle in fixed angular increments.

Item 5

[0158] The robotic surgical system according to any one of items 1 to 4, wherein [0159] the surgical instrument includes an endoscope; and [0160] the robotic surgical system comprises a display to display an image captured by the endoscope in accordance with the coordinate system of the surgical apparatus.

Item 6

[0161] The robotic surgical system according to any one of items 1 to 5, comprising: [0162] a storage to store the predetermined angle received for each operator; wherein [0163] the controller is configured or programmed to: [0164] retrieve the predetermined angle received for the each operator and stored in the storage; and [0165] set the coordinate system of the operation unit based on a retrieved predetermined angle.

Item 7

[0166] The robotic surgical system according to any one of items 1 to 6, wherein [0167] the surgical instrument includes an endoscope; [0168] the operation apparatus includes a monitor to display an image captured by the endoscope and allow a head of the operator to be inserted thereinto; [0169] the monitor has a rotation angle adjusted around an axis along a right-left direction of the operator who operates the operation apparatus; and [0170] the receiver is operable to receive the change in the predetermined angle independently of adjustment of the rotation angle of the monitor.

Item 8

[0171] The robotic surgical system according to any one of items 1 to 7, wherein [0172] the surgical instrument includes an endoscope; [0173] the operation apparatus includes a monitor to display an image captured by the endoscope and allow a head of the operator to be inserted thereinto; [0174] the monitor has a rotation angle adjusted around an axis along a right-left direction of the operator who operates the operation apparatus; [0175] the predetermined angle is changed in conjunction with rotation of the monitor; and [0176] the receiver is operable to receive a further change in the predetermined angle that has been changed in conjunction with the rotation of the monitor.

Item 9

[0177] The robotic surgical system according to any one of items 1 to 8, comprising: [0178] a monitoring controller configured or programmed to monitor command values of the controller for the robot arm and the surgical instrument, and actual axis values of the robot arm and the surgical instrument; wherein [0179] the monitoring controller is configured or programmed to perform a control to issue a warning when differences between the command values of the controller and the actual axis values exceed a predetermined threshold.

Item 10

[0180] The robotic surgical system according to any one of items 1 to 9, wherein [0181] the predetermined angle is an angle by which the coordinate system of the operation unit is rotated in a direction away from the operator with respect to a line perpendicular to a surface on which the operation apparatus is placed; and [0182] the receiver includes a touch panel on the operation apparatus to receive a setting for the robotic surgical system.

Item 11

[0183] The robotic surgical system according to item 10, wherein the receiver is operable to receive the change in the predetermined angle in fixed angular increments.

Item 12

[0184] The robotic surgical system according to any one of items 1 to 11, wherein [0185] the predetermined angle is an angle by which the coordinate system of the operation unit is rotated in a direction away from the operator with respect to a line perpendicular to a surface on which the operation apparatus is placed; [0186] the surgical instrument includes an endoscope; and [0187] the robotic surgical system comprises a display to display an image captured by the endoscope in accordance with the coordinate system of the surgical apparatus.

Item 13

[0188] The robotic surgical system according to item 12, comprising: [0189] a storage to store the predetermined angle received for each operator; wherein [0190] the controller is configured or programmed to: [0191] retrieve the predetermined angle received for the each operator and stored in the storage; and [0192] set the coordinate system of the operation unit based on a retrieved predetermined angle.

Item 14

[0193] The robotic surgical system according to any one of items 11 to 13, wherein [0194] the surgical instrument includes an endoscope; [0195] the robotic surgical system comprises a display to display an image captured by the endoscope in accordance with the coordinate system of the surgical apparatus; [0196] the robotic surgical system comprises a storage to store the predetermined angle received for each operator; and [0197] the controller is configured or

programmed to: [0198] retrieve the predetermined angle received for the each operator and stored in the storage; and [0199] set the coordinate system of the operation unit based on a retrieved predetermined angle.
Item 15
[0200] The robotic surgical system according to item 14, wherein [0201] the surgical instrument includes an endoscope; [0202] the operation apparatus includes a monitor to display an image captured by the endoscope and allow a head of the operator to be inserted thereinto; [0203] the monitor has a rotation angle adjusted around an axis along a right-left direction of the operator who operates the operation apparatus; [0204] the predetermined angle is changed in conjunction with rotation of the monitor; and [0205] the receiver is operable to receive a further change in the predetermined angle that has been changed in conjunction with the rotation of the monitor.
Item 16
[0206] A control method for a robotic surgical system, the control method comprising: [0207] receiving a change by an operator in a predetermined angle of a coordinate system of an operation unit rotated by the predetermined angle with respect to a coordinate system of a surgical apparatus; [0208] changing the predetermined angle of the coordinate system of the operation unit based on a received change in the predetermined angle; [0209] receiving an operation with the operation unit for a surgical instrument attached to a robot arm in a state in which the coordinate system of the operation unit has been changed; and [0210] moving the surgical instrument by the robot arm based on a received operation.
Item 17
[0211] A storage medium operable to store a program for a control method for a robotic surgical system, the control method comprising: [0212] receiving a change by an operator in a predetermined angle of a coordinate system of an operation unit rotated by the predetermined angle with respect to a coordinate system of a surgical apparatus; [0213] changing the predetermined angle of the coordinate system of the operation unit based on a received change in the predetermined angle; [0214] receiving an operation with the operation unit for a surgical instrument attached to a robot arm in a state in which the coordinate system of the operation unit has been changed; and [0215] moving the surgical instrument by the robot arm based on a received operation.

Claims

1. A robotic surgical system comprising: a surgical apparatus including a robot arm to allow a surgical instrument to be attached thereto; an operation apparatus including an operation unit to receive an operation for the surgical instrument, the operation unit having a coordinate system rotated by a predetermined angle with respect to a coordinate system of the surgical apparatus; a controller configured or programmed to perform a control to move the surgical instrument by the robot arm based on the operation received by the operation unit; and a receiver to receive a change in the predetermined angle by an operator.
2. The robotic surgical system according to claim 1, wherein the predetermined angle is an angle by which the coordinate system of the operation unit is rotated in a direction away from the operator with respect to a line perpendicular to a surface on which the operation apparatus is placed.
3. The robotic surgical system according to claim 1, wherein the receiver includes a touch panel on the operation apparatus to receive a setting for the robotic surgical system.
4. The robotic surgical system according to claim 1, wherein the receiver is operable to receive the change in the predetermined angle in fixed angular increments.
5. The robotic surgical system according to claim 1, wherein the surgical instrument includes an endoscope; and the robotic surgical system comprises a display to display an image captured by the endoscope in accordance with the coordinate system of the surgical apparatus.

6. The robotic surgical system according to claim 1, comprising: a storage to store the predetermined angle received for each operator; wherein the controller is configured or programmed to: retrieve the predetermined angle received for the each operator and stored in the storage; and set the coordinate system of the operation unit based on a retrieved predetermined angle.
7. The robotic surgical system according to claim 1, wherein the surgical instrument includes an endoscope; the operation apparatus includes a monitor to display an image captured by the endoscope and allow a head of the operator to be inserted thereinto; the monitor has a rotation angle adjusted around an axis along a right-left direction of the operator who operates the operation apparatus; and the receiver is operable to receive the change in the predetermined angle independently of adjustment of the rotation angle of the monitor.
8. The robotic surgical system according to claim 1, wherein the surgical instrument includes an endoscope; the operation apparatus includes a monitor to display an image captured by the endoscope and allow a head of the operator to be inserted thereinto; the monitor has a rotation angle adjusted around an axis along a right-left direction of the operator who operates the operation apparatus; the predetermined angle is changed in conjunction with rotation of the monitor; and the receiver is operable to receive a further change in the predetermined angle that has been changed in conjunction with the rotation of the monitor.
9. The robotic surgical system according to claim 1, comprising: a monitoring controller configured or programmed to monitor command values of the controller for the robot arm and the surgical instrument, and actual axis values of the robot arm and the surgical instrument; wherein the monitoring controller is configured or programmed to perform a control to issue a warning when differences between the command values of the controller and the actual axis values exceed a predetermined threshold.
10. The robotic surgical system according to claim 1, wherein the predetermined angle is an angle by which the coordinate system of the operation unit is rotated in a direction away from the operator with respect to a line perpendicular to a surface on which the operation apparatus is placed; and the receiver includes a touch panel on the operation apparatus to receive a setting for the robotic surgical system.
11. The robotic surgical system according to claim 10, wherein the receiver is operable to receive the change in the predetermined angle in fixed angular increments.
12. The robotic surgical system according to claim 1, wherein the predetermined angle is an angle by which the coordinate system of the operation unit is rotated in a direction away from the operator with respect to a line perpendicular to a surface on which the operation apparatus is placed; the surgical instrument includes an endoscope; and the robotic surgical system comprises a display to display an image captured by the endoscope in accordance with the coordinate system of the surgical apparatus.
13. The robotic surgical system according to claim 12, comprising: a storage to store the predetermined angle received for each operator; wherein the controller is configured or programmed to: retrieve the predetermined angle received for the each operator and stored in the storage; and set the coordinate system of the operation unit based on a retrieved predetermined angle.
14. The robotic surgical system according to claim 11, wherein the surgical instrument includes an endoscope; the robotic surgical system comprises a display to display an image captured by the endoscope in accordance with the coordinate system of the surgical apparatus; the robotic surgical system comprises a storage to store the predetermined angle received for each operator; and the controller is configured or programmed to: retrieve the predetermined angle received for the each operator and stored in the storage; and set the coordinate system of the operation unit based on a retrieved predetermined angle.
15. The robotic surgical system according to claim 14, wherein the surgical instrument includes an

endoscope; the operation apparatus includes a monitor to display an image captured by the endoscope and allow a head of the operator to be inserted thereinto; the monitor has a rotation angle adjusted around an axis along a right-left direction of the operator who operates the operation apparatus; the predetermined angle is changed in conjunction with rotation of the monitor; and the receiver is operable to receive a further change in the predetermined angle that has been changed in conjunction with the rotation of the monitor.

16. A control method for a robotic surgical system, the control method comprising: receiving a change by an operator in a predetermined angle of a coordinate system of an operation unit rotated by the predetermined angle with respect to a coordinate system of a surgical apparatus; changing the predetermined angle of the coordinate system of the operation unit based on a received change in the predetermined angle; receiving an operation with the operation unit for a surgical instrument attached to a robot arm in a state in which the coordinate system of the operation unit has been changed; and moving the surgical instrument by the robot arm based on a received operation.

17. A storage medium operable to store a program for a control method for a robotic surgical system, the control method comprising: receiving a change by an operator in a predetermined angle of a coordinate system of an operation unit rotated by the predetermined angle with respect to a coordinate system of a surgical apparatus; changing the predetermined angle of the coordinate system of the operation unit based on a received change in the predetermined angle; receiving an operation with the operation unit for a surgical instrument attached to a robot arm in a state in which the coordinate system of the operation unit has been changed; and moving the surgical instrument by the robot arm based on a received operation.